

SMART URBAN MOBILITY & TRAFFIC INTELLIGENCE DASHBOARD

Milestone 1 — Data Acquisition, Preprocessing & Dimensional Modeling

Abstract

This milestone establishes the data foundation for a Smart Urban Mobility and Traffic Intelligence Dashboard. Mobility, transit, weather, and demographic information was consolidated, cleaned through an ETL workflow, reduced to an analysis-ready dataset, and structured as a star schema in Power BI. The final model contains 500 records and 11 retained fields, with FactTraffic as the central fact table and DimDate, DimRoute, DimLocation, and DimWeather as supporting dimensions. The completed Power BI report provides KPI cards, trend and comparison visuals, route analysis, and interactive slicers for date, route, and weather.

1. Introduction

Urban mobility data can contain information from several domains, including traffic movement, passenger activity, weather, and population. Combining these sources in a structured business-intelligence model makes it easier to monitor mobility patterns and compare route performance.

The objective of Milestone 1 was to prepare this analytical foundation. The work therefore focused on four activities: acquiring and consolidating the required data, cleansing and transforming the source data, designing a star schema, and loading and validating the model in Power BI.

Milestone 1 Objectives

- Acquire and consolidate mobility, transit, weather, and demographic data.
- Develop an ETL and data-cleansing workflow using Power Query.
- Design a star schema with one fact table and four dimension tables.
- Load the model into Power BI and validate relationships and totals.
- Create the initial dashboard with KPIs, charts, and slicers.

Project Data Scope

Stage	Rows	Columns	Purpose
Raw source	2,000	28	Initial dataset containing relevant and non-essential fields
Analysis-ready	500	11	Focused dataset retained for the dashboard

The retained fields cover Date, RouteID, Location, Passenger Count, Average Speed, Vehicle Count, Weather Condition, Rainfall, Temperature, Population, and Area km².

2. ETL & DATA CLEANING

2.1 Data Preparation

The source dataset originally contained 2,000 records and 28 attributes. During transformation, redundant or non-essential fields were removed and the working dataset was reduced to 500 records and 11 fields. Date values were standardised; numeric fields were checked for valid numeric values; and categorical weather values were kept in a consistent set: Clear, Rain, Snow, and Fog.

2.2 Power Query (M) Formulas

The following compact M expressions represent the main transformations documented for the milestone: selecting the required columns, retaining the 500-row working dataset, applying data types, removing duplicates, and creating calendar attributes.

```
= Table.SelectColumns(Source, {"Date","RouteID","Location","Passenger Count","Average Speed","Vehicle Count","Weather Condition","Rainfall","Temperature","Population","Area km²"})
= Table.FirstN(PreviousStep, 500)
= Table.TransformColumnTypes(PreviousStep, {"Date", type date}, {"RouteID", type text}, {"Location", type text}, {"Passenger Count", Int64.Type}, {"Average Speed", type number}, {"Vehicle Count", Int64.Type}, {"Weather Condition", type text}, {"Rainfall", type number}, {"Temperature", type number}, {"Population", Int64.Type}, {"Area km²", type number})
= Table.Distinct(PreviousStep)
= Table.AddColumn(PreviousStep, "Year", each Date.Year([Date]), Int64.Type)
= Table.AddColumn(PreviousStep, "Month", each Date.Month([Date]), Int64.Type)
= Table.AddColumn(PreviousStep, "Day Name", each Date.DayOfWeekName([Date]), type text)
```

2.3 ETL Summary

ETL Step	Action	Result
Extract	Import source mobility and related data	Raw 2,000 × 28 dataset
Transform	Select required fields and clean data	11 relevant fields
Transform	Standardise types and date attributes	Analysis-ready values
Transform	Remove duplicate/invalid records as applicable	Cleaner model
Load	Import into Power BI and build relationships	Star-schema model

2.4 Data Types & Validation

Dates are stored as date values, identifiers as text, counts and population as whole numbers, and measurements such as speed, rainfall, temperature, and area as numeric values. Validation included checking the 500 fact records, relationship keys, aggregate totals, and filtering behaviour in Power BI.

3. DATA MODEL & DAX MEASURES

3.1 Star Schema Design

The cleaned dataset was organised into a star schema. FactTraffic stores the measurable traffic observations and keys, while four dimension tables provide descriptive context for filtering and grouping.

Table	Main Fields / Role
FactTraffic	Date, RouteID, Location, Weather Condition, Passenger Count, Vehicle Count, Average Speed, Rainfall, Temperature, Population, Area km²
DimDate	Date, Day Name, Month, Month Number, Year
DimRoute	Route ID
DimLocation	Location
DimWeather	Weather

Relationships are one-to-many from each dimension to FactTraffic: DimDate → FactTraffic, DimRoute → FactTraffic, DimLocation → FactTraffic, and DimWeather → FactTraffic.

3.2 DAX Measures

The main dashboard KPIs are calculated with the following DAX measures:

Total Passengers = SUM(FactTraffic[Passenger Count])

Average Speed = AVERAGE(FactTraffic[Average Speed])

Total Population = SUM(FactTraffic[Population])

Average Rainfall = AVERAGE(FactTraffic[Rainfall])

Measure	DAX Function	Dashboard Use
Total Passengers	SUM	Total recorded passenger volume
Average Speed	AVERAGE	Overall traffic speed in km/h
Total Population	SUM	Combined population represented in the data
Average Rainfall	AVERAGE	Average recorded rainfall in mm

3.3 Dashboard KPIs

KPI	Displayed Value
Total Passengers	64K
Average Speed (km/h)	42.37
Total Population	285M
Average Rainfall (mm)	3

3.4 Key Validation Checks

- FactTraffic contains the expected 500 records after loading.
- Dimension keys match the corresponding fact-table values.
- One-to-many relationships are correctly represented in the model.
- KPI totals were cross-checked with the source/cleaned dataset.
- Route, weather, and date slicers affect the dashboard visuals through the model relationships.

4. STAR SCHEMA IMPLEMENTATION

Figure 1 shows the Power BI model view used for Milestone 1. FactTraffic is positioned at the centre and is connected to the four dimension tables used for analysis and filtering.

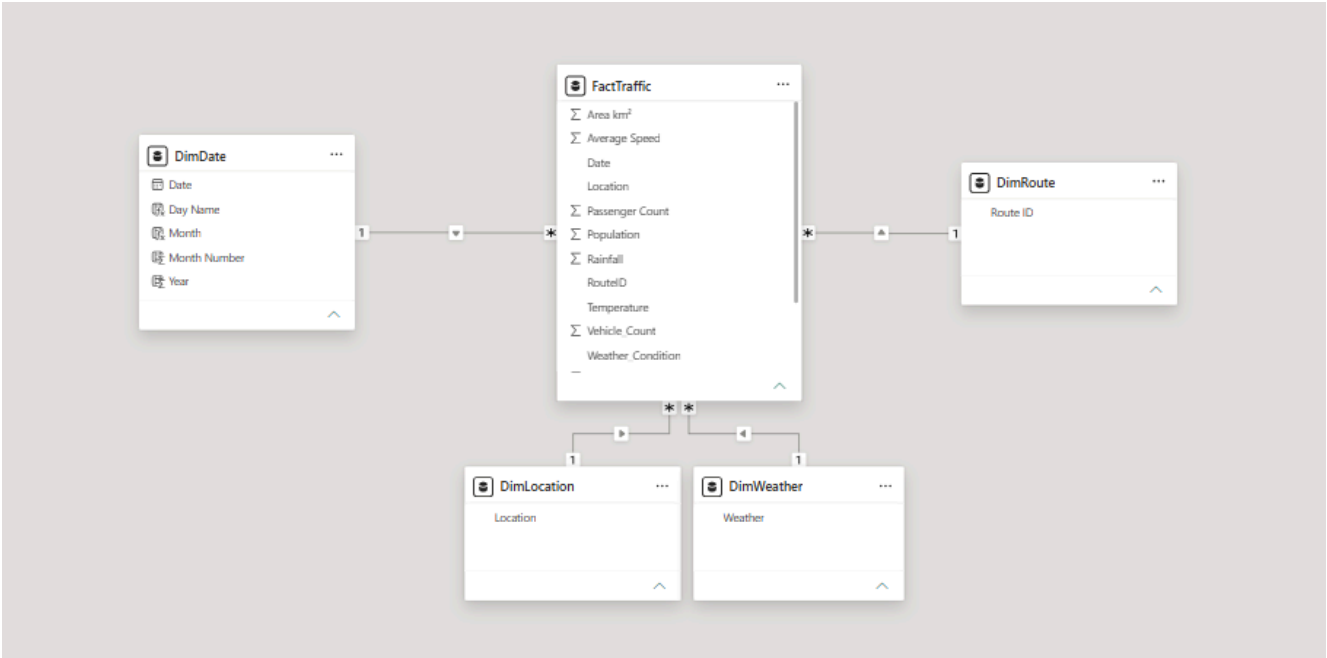


Figure 1: Star schema showing FactTraffic connected to DimDate, DimRoute, DimLocation, and DimWeather.

Model Interpretation

DimDate supports time-based analysis through Date, Day Name, Month, Month Number, and Year. DimRoute identifies individual routes, DimLocation provides the service-area context, and DimWeather provides the weather categories. FactTraffic contains the quantitative observations used by the dashboard measures.

Relationship Structure

Dimension	Fact Table	Cardinality
DimDate	FactTraffic	1 : Many
DimRoute	FactTraffic	1 : Many
DimLocation	FactTraffic	1 : Many
DimWeather	FactTraffic	1 : Many

This structure reduces repeated descriptive data and allows the same fact measures to be analysed consistently by date, route, location, and weather.

5. DASHBOARD RESULTS & CONCLUSION

Figure 2 presents the completed Milestone 1 Power BI dashboard. It combines four KPI cards, analytical visuals, and slicers in a single interactive report page.

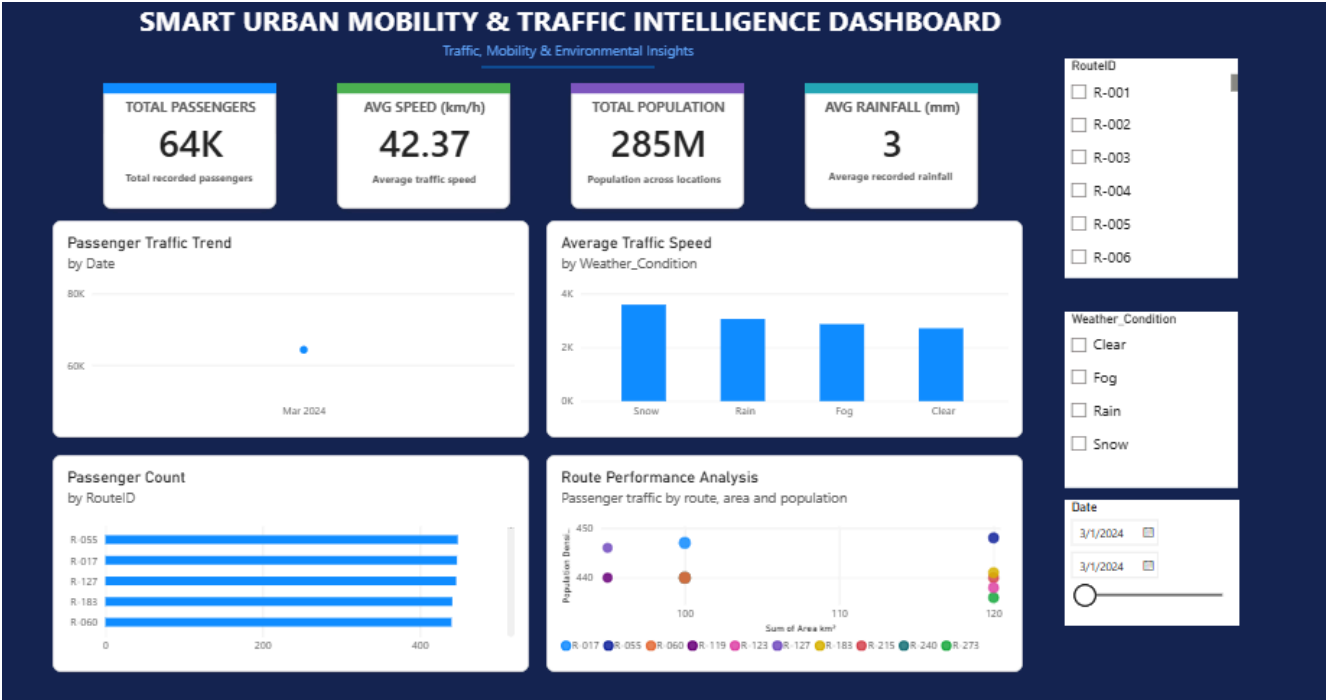


Figure 2: Smart Urban Mobility & Traffic Intelligence Dashboard — Power BI report view.

5.1 Dashboard Visuals

Visual	Purpose
Passenger Traffic Trend	Shows passenger traffic over time.
Average Traffic Speed by Weather	Compares average speed across Snow, Rain, Fog, and Clear.
Passenger Count by Route	Ranks routes by total passenger count.
Route Performance Analysis	Compares route area and population density context.

The dashboard also contains RouteID, Weather Condition, and Date slicers. These controls allow the user to move from an overall view to a specific route, weather condition, or time period.